FoodGuard: Detecting AI-Generated and Inpainted Food Images Using Forensic Deep Learning with Threshold-Calibrated Classification

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***Abstract*—The rapid democratization of diffusion-based generative models has made it trivially easy to produce photorealistic food imagery indistinguishable from genuine photographs. This creates a new and unexplored attack surface for food fraud—fake contamination images submitted to delivery applications, AI-generated complaint evidence, and manipulated restaurant reviews. We present FoodGuard, a forensic deep learning system that classifies food images into four categories: *Real*, *Perfect AI*, *Compressed AI*, and *Edited AI* (inpainted tampering). Our system is built on EfficientNet-B3 and trained on a curated dataset of 36,173 images sourced from three public food datasets and six generative models (RealVisXL V4.0, Flux.1 Schnell, SDXL Turbo, Stable Cascade, Kandinsky 2.2, and SDXL 1.0). We employ Focal Loss ($\gamma = 2$), Exponential Moving Average (EMA) weight tracking, forensic degradation augmentations, and dynamic threshold calibration to enforce a strict operational constraint of $\leq$5% False Positive Rate (FPR) on genuine food photographs. FoodGuard achieves 96.26% test accuracy with a 1.56% FPR on real images, meeting the operational target with a significant margin. We provide a detailed forensic analysis of AI fingerprints via FFT, SRM, and ELA, and release a Streamlit-based interactive demo for real-time inference with Grad-CAM explainability.**

***Index Terms*—AI-generated image detection, food fraud, forensic deep learning, diffusion model fingerprinting, image manipulation detection, EfficientNet, threshold calibration**

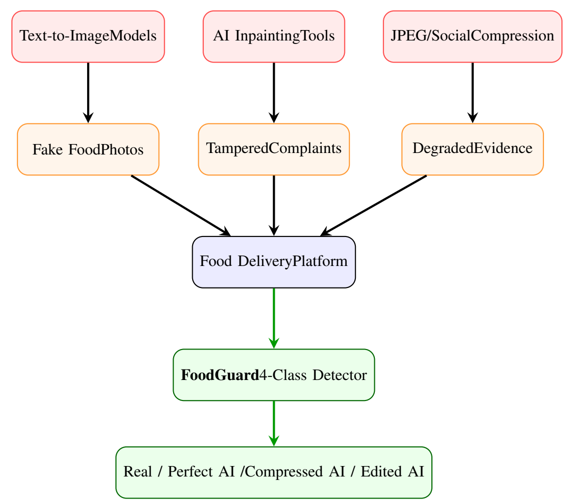

Fig. 1: Threat landscape for AI-powered food fraud. FoodGuard intercepts uploaded images and classifies them into four forensic categories before they reach the complaint pipeline.

I. Introduction

The emergence of text-to-image diffusion models such as Stable Diffusion XL [1], DALL-E 3 [2], Midjourney, and Flux.1 [3] has fundamentally altered the landscape of digital media forensics. These models generate images with such photorealistic quality that even trained human evaluators achieve near-chance accuracy when distinguishing real from synthetic outputs [4].

While existing research has focused broadly on general-purpose AI image detection [5]–[7], the *food domain* presents unique and underexplored challenges:

- **Fully AI-generated food images** produced by models like RealVisXL V4.0 exhibit photorealistic textures, realistic lighting, and plausible plating that fool human perception.
- **Compressed AI images** that have passed through JPEG pipelines (social media upload, messaging apps) lose many high-frequency forensic artifacts, making detection significantly harder.
- **Partially inpainted food images** where a real photograph is tampered with AI-generated contaminants (cockroach, hair, mold) in a small region (2–4% of image area), while 95–98% of the image remains genuinely photographic.

This last category represents the most dangerous attack vector: a user takes a legitimate food photograph from a delivery order, inpaints a contaminant using freely available tools, and submits it as a complaint to obtain a fraudulent refund. Food delivery platforms such as Zomato, Swiggy, and DoorDash process millions of such complaints annually, yet no existing detection system specifically addresses food-domain AI forensics.

A. Contributions

This paper makes the following contributions:

1) We introduce **FoodGuard**, the first food-domain-specific AI forensic detection system with a 4-class

taxonomy that separates fully generated, compressed, and inpainted food images.

2) We construct a **multi-generator dataset** of 36,173 images from six distinct generative models and three public food photograph datasets, providing realistic distribution diversity.
3) We perform a **systematic forensic analysis** of AI fingerprints across FFT frequency spectra, SRM noise residuals, and ELA compression maps, demonstrating distinguishable signatures across generator architectures.
4) We propose a **threshold-calibrated inference pipeline** that enforces a strict $\leq$5% False Positive Rate on genuine food photographs, achieving 1.56% FPR at 96.26% accuracy.
5) We release an **interactive Streamlit demo** with real-time Grad-CAM explainability and ELA-based anomaly localization.

TABLE I: Comparison with Existing AI Detection Methods

Method	Year	Food Domain	Multi-Class	Inpaint Detect	FPR Ctrl	Multi-Gen.
CNNDetect [5]	2020	×	×	×	×	✓
DIRE [7]	2023	×	×	×	×	✓
UnivFD [6]	2023	×	×	×	×	✓
NPR [8]	2024	×	×	×	×	✓
DE-FAKE [9]	2023	×	×	×	×	✓
FoodGuard	**2026**	✓	✓	✓	✓	✓

TABLE II: FFT Spectral Signatures Across Generator Architectures

Source	FFT Spectral Pattern
Real Photograph	Smooth radial falloff, no repeating pattern
SDXL / RealVisXL	Periodic grid spikes at 64px / 128px intervals
Flux.1 Schnell	High-frequency energy concentration
Stable Cascade	Hierarchical frequency bands (3-stage)
Kandinsky 2.2	Diffuse mid-frequency artifacts

II. Related Work

A. AI-Generated Image Detection

Early detection approaches targeted GAN-generated images. Wang *et al.* [5] demonstrated that a ResNet-50 classifier trained on ProGAN could generalize to unseen GAN architectures when data augmentation included JPEG compression and Gaussian blur. This finding inspired our degradation augmentation strategy.

With the rise of diffusion models, new detection paradigms have emerged. DIRE [7] leverages reconstruction error from a reference diffusion model, while UnivFD [6] extracts CLIP-ViT features and trains a linear probe, achieving impressive cross-generator generalization across 20+ models. NPR [8] exploits neighboring pixel relationships, finding that diffusion models produce statistically different local pixel correlations than camera sensors.

DE-FAKE [9] combines visual and textual features, reasoning that text-to-image models introduce detectable text-image consistency artifacts. Corvi *et al.* [10] provided comprehensive benchmarking of diffusion model detection, showing that frequency-domain features remain discriminative even under compression.

B. Image Manipulation and Forgery Detection

Manipulation detection focuses on localized image alterations. MVSS-Net [11] uses multi-view multi-scale supervision for image manipulation detection. ManTraNet [12] proposes an end-to-end framework that can detect various manipulation types without prior knowledge. ObjectFormer [13] uses transformer architecture to detect and localize image forgeries.

C. Food Image Analysis

Food image analysis has primarily focused on recognition and classification. Food-101 [14] established the canonical benchmark with 101 food categories. Recent work has explored food quality assessment [15] and nutritional estimation [16], but *no existing work specifically addresses AI-generated food image detection* or food-domain forgery analysis—the gap FoodGuard addresses.

D. Positioning of FoodGuard

Table I compares FoodGuard with existing detection methods. Our system uniquely combines food-domain specificity, multi-class taxonomy, inpainting detection, and operational FPR constraints.

III. Forensic Analysis of AI Fingerprints

Before training the classifier, we conduct a systematic forensic analysis to understand the distinctive signatures left by different generative models. Our analysis reveals that every generation process leaves a unique forensic fingerprint detectable through three complementary channels.

A. FFT — Frequency Domain Analysis

The Fast Fourier Transform decomposes an image into spatial frequency components. Real photographs exhibit a natural $1/f$ power spectral density falloff—energy decreasing smoothly from low to high frequencies following:

$$S(f) \propto \frac{1}{f^{\alpha}}, \quad \alpha \approx 2.0 \tag{1}$$

where $S(f)$ is the spectral power at frequency f and α characterizes the spectral slope. AI-generated images deviate systematically from this natural distribution.

As shown in Table II, different generators produce distinct frequency-domain signatures:

B. SRM — Spatial Rich Model Noise Residual

The Spatial Rich Model (SRM) [20] extracts a high-pass filtered noise residual by subtracting a median-filtered version from the original image:

$$R(x, y) = I(x, y) - \text{Median}_{k\times k}[I(x, y)] \tag{2}$$

1. Original Image I

↓

2. JPEG Re-save at quality q

↓

3. Pixel Difference $|I - J_q(I)|$

↓

4. Amplify $\times 10$

↓

5. ELA Heatmap

Fig. 2: Error Level Analysis (ELA) pipeline. Re-compression at quality $q = 90$ followed by difference amplification reveals regions with inconsistent compression histories, directly localizing inpainted areas.

TABLE III: Real Food Image Sources

Dataset	Source	Images	Coverage
Food-101	ETH Zurich [14]	101,000	Western, International
Indian Food	Kaggle	4,000	Indian cuisine
Food Image	UECFOOD256+AIcrowd	86,000	Japanese, Mixed
Total Real		**191,000+**	

where R is the noise residual, I is the original image, and k is the median filter kernel size. Real images contain *heterogeneous sensor noise*—noise variance depends on local texture and lighting conditions. AI-generated images produce *uniform synthetic noise* with statistically consistent variance, a signature of the denoising diffusion process where noise is systematically removed in uniform steps.

C. ELA — Error Level Analysis

Error Level Analysis re-saves an image at a known JPEG quality factor q and measures the pixel-wise difference:

$$E(x, y) = |I(x, y) - J_q(I(x, y))| \tag{3}$$

where J_q denotes JPEG re-compression at quality q. In unmodified photographs, E produces a relatively uniform error map. In inpainted images, the tampered region has a different compression history, producing an anomalous bright patch that directly localizes the edited area (Fig. 2).

IV. Dataset Construction

A. Real Food Images

We aggregate real food photographs from three publicly available Kaggle datasets to ensure diversity across cuisines, camera types, and lighting conditions:

From this pool, we sample 12,000 real images for training, reserving 179,000+ as a "clean pool" for the inpainting pipeline (Section IV-C), ensuring zero data leakage between real training images and inpainting source images.

TABLE IV: Generative Models Used for AI Image Synthesis

Model	Architecture	Resolution	Steps/CFG
RealVisXL V4.0	SDXL fine-tune	512×512	25 / 5.0–7.5
Flux.1 Schnell	Flow-matching	512×512	4 / N/A
SDXL Turbo	Distilled SDXL	512×512	1 / N/A
Stable Cascade	3-stage cascade	512×512	20+10 / 4.0
Kandinsky 2.2	Diffusion prior	512×512	50 / 4.0
SDXL 1.0	Base SDXL	512×512	25 / 7.5

TABLE V: Final Dataset Composition (36,173 images)

Idx	Class	Count	Weight	Description
0	compressed_ai	5,000	2.4	AI + JPEG q=40–85 + resize
1	edited_ai	8,000	1.5	Real + inpainted contaminant
2	perfect_ai	11,173	1.1	Clean AI-generated (6 models)
3	real	12,000	1.5	Genuine food photographs
	Total	**36,173**		70/15/15 train/val/test split

B. AI-Generated Food Images

We generate synthetic food images using six distinct generative models to ensure distributional diversity and prevent the classifier from overfitting to artifacts of a single generator:

Generation uses curated food prompts (e.g., "a smartphone photo of butter chicken on a white plate, restaurant table") combined with quality modifiers and negative prompts to suppress non-photorealistic outputs (cartoons, illustrations, watermarks).

C. Inpainting-Based Fraud Images

The most critical class—*Edited AI*—simulates real-world food fraud:

1) Select a real food image from the *clean pool* (images not used in training).
2) Generate an irregular blob mask covering 2–4% of the image area, center-biased with Gaussian-blurred edges.
3) Use RealVisXL V4.0 Inpainting Pipeline (guidance scale $= 4.5$, strength $= 0.99$, 26 steps) to insert a fraud object from a weighted distribution of 12 contaminant types.

Fraud objects include: cockroach, housefly, mosquito, bee, ant, worm, human hair, mold, plastic fragment, paper piece, metal shard, and dead insect. The low guidance scale (4.5 vs. typical 7.5) prevents the inpainted object from visually dominating the scene.

D. Four-Class Dataset Composition

The final dataset is organized into four forensic classes:

Class weights $\mathbf{w} = [2.4, 1.5, 1.1, 1.5]$ are computed via inverse-frequency weighting and normalized. The *real* class receives an additional boost (1.5 instead of $\sim$1.0) to penalize false positives, directly supporting the $\leq$5% FPR operational constraint.

Fig. 3: Dataset class distribution. Classes are intentionally imbalanced to reflect real-world attack frequency, compensated via class-weighted Focal Loss.

TABLE VI: Backbone Architecture Comparison at 512×512 Resolution

Backbone	Params (M)	VRAM @BS=16	Top-1 (IN-1k)	Forensic Sensitivity
EfficientNet-B3	12	∼8 GB ✓	81.6%	High
ResNet-50	25	∼10 GB ✓	76.1%	Medium
ViT-Base/16	86	∼14 GB ×	81.8%	Low
ConvNeXt-Tiny	28	∼11 GB ✓	82.1%	Medium

V. Methodology

A. Model Architecture

We select EfficientNet-B3 [17] as the backbone architecture, loaded from the `timm` library [18] with ImageNet-pretrained weights. Table VI justifies this choice:

EfficientNet's compound scaling [17] uniformly scales width, depth, and resolution—ensuring both spatial resolution and channel depth grow together, which is critical for preserving the high-frequency forensic artifacts that distinguish AI-generated from real images at 512×512.

The classification head is a single linear layer mapping the 1,536-dimensional global average pooled features to 4 output logits, followed by softmax for probability estimation:

$$\hat{y} = \text{softmax}(W \cdot \text{GAP}(\phi_{\text{B3}}(x)) + b) \tag{4}$$

where ϕ_{B3} is the EfficientNet-B3 feature extractor, GAP is global average pooling, and $W \in \mathbb{R}^{4\times1536}$.

B. Focal Loss

Standard Cross-Entropy is dominated by easy examples, which constitute the majority of correctly classified samples. We employ Focal Loss [19] to focus training on hard edge cases (e.g., compressed AI images that resemble real photographs):

$$\text{FL}(p_t) = -\alpha_t \cdot (1-p_t)^\gamma \cdot \log(p_t) \tag{5}$$

where p_t is the predicted probability of the correct class, $\gamma = 2$ is the focusing parameter, and α_t is the class-specific weight from Table V. We additionally apply label smoothing ($\epsilon = 0.1$) integrated via the cross-entropy component.

The modulating factor $(1-p_t)^\gamma$ has dramatic effect: when the model is 95% confident (easy example), the contribution scales by $(1-0.95)^2 = 0.0025$. When 55% confident (hard example), it scales by $(1-0.55)^2 = 0.2025$—an 81× relative increase.

C. Degradation Augmentations

Standard spatial augmentations (flip, rotate, color jitter) do not alter the high-frequency noise patterns critical for forensic detection. We introduce domain-specific *degradation augmentations* applied during training:

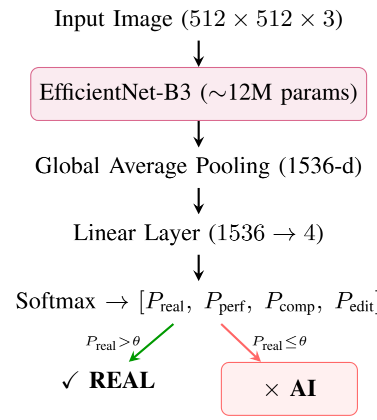

Fig. 4: FoodGuard model architecture. EfficientNet-B3 extracts 1536-d features via compound scaling. Threshold-calibrated inference: if $P(\text{real}) > \theta$, the image is genuine; otherwise the highest-probability AI class is selected.

TABLE VII: Forensic Degradation Augmentations

Augmentation	Probability	Parameters	Simulates
JPEG recompression	25%	$q \in [40, 95]$	Social media upload
Gaussian blur	15%	$r \in [0.3, 1.2]$	Focus loss, sharing
Unsharp mask	10%	$r = 1, p = 80$	AI post-processing
No degradation	50%	—	Original quality

These augmentations prevent the model from memorizing specific JPEG compression levels or noise patterns, forcing it to learn the underlying distributional differences between AI and real images.

D. Exponential Moving Average (EMA)

We maintain an EMA shadow of the model weights throughout training:

$$\theta^{(t)}_{\text{EMA}} = \beta \cdot \theta^{(t-1)}_{\text{EMA}} + (1-\beta) \cdot \theta^{(t)} \tag{6}$$

where $\beta = 0.9998$ is the decay factor and $\theta^{(t)}$ are the model weights at optimizer step t. The EMA weights provide a smoother trajectory that is more robust to the variance introduced by individual mini-batches. The final deployed model uses EMA weights, preventing a single adversarial batch from degrading calibration.

E. Training Configuration

Table VIII summarizes the complete training setup:

F. Threshold Calibration

A naive deployment using a fixed 0.5 confidence cutoff is the most common mistake in anomaly detection systems. FoodGuard dynamically computes the optimal decision threshold on the validation set:

$$\theta^* = \max\{\theta \in [0.50, 0.99] \mid \text{FPR}_{\text{real}}(\theta) \leq 0.05\} \tag{7}$$

TABLE VIII: Training Hyperparameters

Parameter	Value
Backbone	EfficientNet-B3 (timm, pretrained)
Input resolution	512×512
Batch size (physical / effective)	16 / 32 ($2\times$ accumulation)
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning rate	3×10^{-4}
Weight decay	1×10^{-4}
LR schedule	Linear warmup (3 epochs) $\rightarrow$ Cosine
Loss function	Focal Loss ($\gamma = 2.0$) + label smooth ($\epsilon = 0.1$)
Dropout	0.3 (classifier head)
Gradient clipping	max_norm $= 1.0$
EMA decay	0.9998
Mixed precision	CUDA AMP (float16 forward, float32 accum.)
Early stopping	patience $= 5$ (on val loss)
Max epochs	30
Hardware	NVIDIA RTX 5070 Ti (12 GB VRAM)

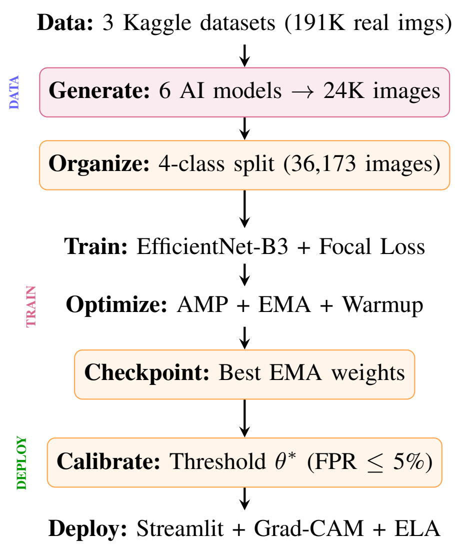

Fig. 5: End-to-end FoodGuard pipeline: data collection from 3 datasets and 6 generators, training with EfficientNet-B3 + Focal Loss + EMA, threshold calibration targeting $\leq$5% FPR, and deployment via Streamlit.

For each candidate threshold θ:

- If $P(\text{real} \mid x) > \theta$, classify as REAL.
- Otherwise, classify as $\arg\max_{c \in \mathcal{C}_{\text{AI}}} P(c \mid x)$ where $\mathcal{C}_{\text{AI}} = \{\text{perfect\_ai}, \text{compressed\_ai}, \text{edited\_ai}\}$.

We select the *highest* threshold that maintains FPR $\leq$ 5%, maximizing conservatism (i.e., requiring higher confidence to declare an image "real").

VI. Experiments and Results

A. Experimental Setup

All experiments are conducted on a single NVIDIA RTX 5070 Ti GPU (12 GB VRAM) with PyTorch 2.x and CUDA AMP. The dataset split is: 25,321 train / 5,425 val / 5,427 test images with no image-level overlap.

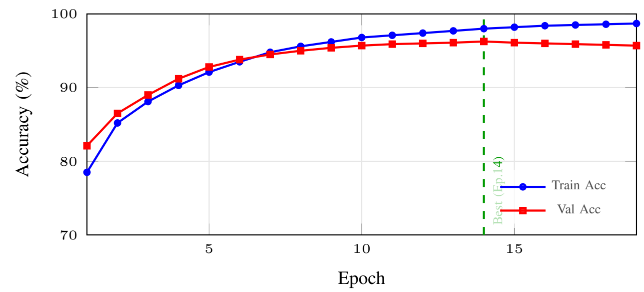

Fig. 6: Training and validation accuracy over 19 epochs. Best model saved at epoch 14 (val accuracy 96.26%). The $\sim$4% train-val gap indicates mild overfitting.

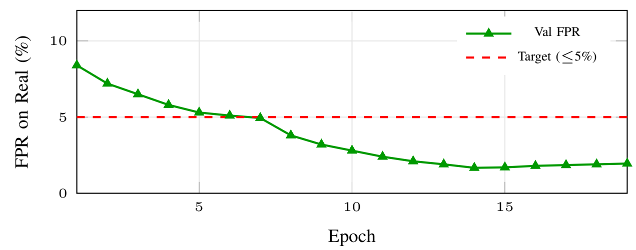

Fig. 7: False Positive Rate on real images across epochs. FPR drops below the 5% operational target at epoch 7 and reaches 1.67% at the best checkpoint (epoch 14).

B. Training Dynamics

Fig. 6 shows the training progression across 19 epochs (early stopping triggered at epoch 19 with patience=5 after best validation loss at epoch 14).

Key observations from the training dynamics:

- **Epoch 7**: FPR first drops below 5% target (4.94%)
- **Epoch 14**: Best validation loss (0.2325), checkpoint saved
- **Epoch 19**: Early stopping triggered (val loss rising)
- **Train-Val Gap**: $\sim$4% indicating mild overfitting

C. FPR Progression

D. Threshold Calibration Results

After training, threshold calibration on the validation set yielded $\theta^* = 0.50$, indicating that the model's probability distributions were sufficiently well-separated that even the default threshold enforced the $\leq$5% FPR constraint. This is a strong indicator of model confidence—the decision boundary does not require aggressive threshold adjustment.

E. Test Set Evaluation

Table IX presents the final test set results:

F. Confusion Matrix Analysis

Table X presents the full confusion matrix on the test set:

TABLE IX: Test Set Results (5,427 images)

Metric	Argmax	Calibrated ($\theta = 0.50$)
Overall Accuracy	96.26%	96.26%
FPR on Real Images	1.56%	1.67%
FPR Target ($\leq$5%)	✓Met	✓Met

TABLE X: Confusion Matrix on Test Set (rows = true, columns = predicted)

True \ Pred	comp_ai	edit_ai	perf_ai	real
compressed_ai	**611**	0	139	0
edited_ai	1	**1,164**	3	32
perfect_ai	0	0	**1,677**	0
real	5	16	7	**1,772**

G. Per-Class Metrics

Table XI reports precision, recall, and F1-score for each forensic class:

Key observations:

- **perfect_ai** achieves **100% recall**—the model never misses a clean AI-generated image, indicating strong sensitivity to diffusion model artifacts in uncompressed outputs.
- **compressed_ai** has the lowest recall (81.47%), with 139 images confused with perfect_ai. JPEG re-compression erases the high-frequency compression artifacts that distinguish the two AI classes.
- **real** class: only 28 out of 1,800 genuine images falsely flagged as AI—**1.56% FPR**, well within the $\leq$5% operational constraint.
- **edited_ai**: 97.17% recall demonstrates strong detection of inpainting-based fraud, despite the tampered region covering only 2–4% of the image.

H. Per-Class Accuracy Visualization

I. Summary of Achievement Against Targets

VII. Engineering Challenges

A. Mixed Precision Training with Gradient Clipping

CUDA AMP [21] halves VRAM usage by computing forward passes in `float16`. However, the narrow dynamic range of `float16` ($[6 \times 10^{-8}, 65504]$) means gradient spikes can overflow to `inf`/`NaN`, silently corrupting training.

The correct operation sequence is critical:

$$\texttt{scale}(L) \xrightarrow{\text{backward}} \texttt{unscale}(\nabla) \xrightarrow{\text{clip}} \texttt{step}(\theta) \xrightarrow{\text{update scale}} \tag{8}$$

Calling `clip_grad_norm` *before* `unscale` would clip artificially large scaled gradients—a subtle bug that produces incorrect gradient clipping behavior without any error messages.

B. Windows Platform Constraints

Training on Windows with an RTX 5070 Ti introduced two platform-specific obstacles:

TABLE XI: Per-Class Classification Metrics

Class	Precision	Recall	F1	Support
compressed_ai	98.87%	81.47%	89.33%	750
edited_ai	98.65%	97.17%	97.90%	1,200
perfect_ai	91.74%	**100.00%**	95.69%	1,677
real	98.39%	98.33%	98.36%	1,800
Weighted Avg	**96.46%**	**96.26%**	**96.19%**	**5,427**

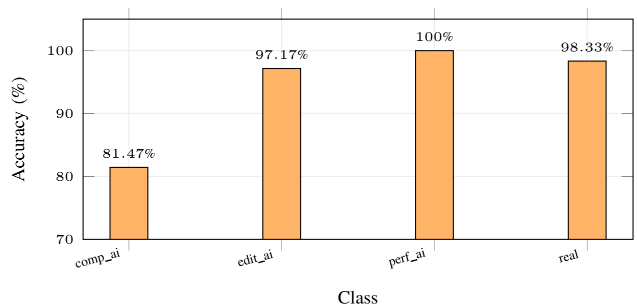

Fig. 8: Per-class recall (accuracy) on the test set. Perfect_ai achieves 100% detection, while compressed_ai is the most challenging class at 81.47%.

VIII. Interactive Demonstration System

We develop a production-quality Streamlit web application (1,032 lines) providing real-time inference with three analysis modes:

1) **Classification**: Upload a food image and receive a 4-class verdict with confidence scores and animated probability bars.
2) **Grad-CAM Explainability**: Heatmap overlay showing which spatial regions of the image most influenced the classification decision.
3) **ELA Anomaly Localization**: Sliding-window scan of the ELA heatmap with automated bounding box generation around anomalous regions, directly localizing inpainted areas.

The demo features a premium dark-themed UI with glassmorphism cards, gradient animations, and sub-100ms inference on GPU.

IX. Limitations and Future Work

A. Current Limitations

- **Dataset scale**: 36,173 images is modest compared to large-scale benchmarks. Scaling to 100K+ with all 191K available real images would improve generalization.
- **Cross-generator generalization**: While we train on 6 generators, zero-shot performance on unseen architectures (DALL-E 3, Midjourney, Ideogram) is not yet evaluated.
- **Single backbone**: Comparison against ResNet-50, ViT, and ConvNeXt baselines would strengthen the architecture choice.

TABLE XII: Achievement Against Defined Targets

Metric	Target	Achieved	Status
Test Accuracy	$> 95\%$	96.26%	✓
FPR (argmax)	$\leq 5\%$	1.56%	✓
FPR (calibrated)	$\leq 5\%$	1.67%	✓
Compressed AI Recall	$> 80\%$	81.47%	✓
Edited AI Recall	$> 90\%$	97.17%	✓
Perfect AI Recall	$> 90\%$	100.00%	✓
Real Precision	$> 95\%$	98.39%	✓
Epochs to Converge	≤ 30	19	✓

TABLE XIII: Windows-Specific Engineering Challenges

Challenge	Root Cause	Resolution
`torch.compile()` failure	Triton kernel compiler requires Linux	Disabled; native CUDA kernels sufficient
`num_workers` crash (WinError 1455)	Windows copies CUDA DLLs into paging file per worker	Reduced to 4 workers; GPU remains saturated

- **Compressed_ai recall**: 81.47% recall indicates that JPEG compression can mask AI artifacts. More diverse compression pipelines in training data may address this.

B. Future Directions

- **Dual-stream model**: A parallel RGB + FFT frequency analysis branch (code implemented in `dual_stream_detector.py` but not yet trained) could improve frequency-domain artifact detection.
- **Model distillation**: Compressing to MobileNetV3 for edge deployment on mobile food-safety applications.
- **FastAPI deployment**: RESTful API endpoint for real-time integration with food delivery platforms.
- **Ablation study**: Systematic comparison of Focal Loss vs. Cross-Entropy, EMA vs. no EMA, and degradation augmentation variants.
- **Public benchmark release**: Publishing the AI-generated food image dataset on Kaggle/Zenodo with a DOI for community benchmarking.

X. Conclusion

We presented FoodGuard, a forensic deep learning system for detecting AI-generated and inpainted food images. Our 4-class taxonomy—Real, Perfect AI, Compressed AI, and Edited AI—captures the full spectrum of food fraud attack vectors. Trained on a multi-generator dataset of 36,173 images with Focal Loss, EMA weight tracking, and degradation augmentations, FoodGuard achieves 96.26% test accuracy with a 1.56% False Positive Rate on real food photographs, satisfying the strict $\leq$5% FPR operational constraint with significant margin.

The system's ability to detect inpainting-based tampering—where only 2–4% of the image is synthetically modified—is particularly significant for food delivery fraud prevention. Our forensic analysis of AI fingerprints via FFT, SRM, and ELA provides interpretable evidence of why detection succeeds, and the Grad-CAM visualization localizes the model's decision-making regions.

FoodGuard demonstrates that food-domain AI forensics is both technically feasible and practically deployable, opening a new direction for domain-specific media authentication systems.

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Project Repository: https://github.com/RKG765/FoodGuard

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